

Ling Xia

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EDUCATION

Princeton University

Graduate Program in Geosciences
Department of Geosciences

Princeton, NJ

Sep. 2024 - Present

University of Science and Technology of China (USTC)

B.S. in Geophysics
School of the Gifted Young (GPA: 4.05/4.30, rank: 1/61)

Hefei, Anhui

Sep. 2020 - Jul. 2024

PUBLICATIONS

1. **Xia, L.** and Rubin, A.M., 2026. A Simple Discrete Elastic Model Exhibits Both the Statistical Properties and Migration Patterns of Slow Slip. *submitted to JGR Solid Earth*.
<https://doi.org/10.22541/essoar.177249008.89007734/v1>

CONFERENCE AND TALKS

3. **Xia, L.** and Rubin, A.M. A Discrete-Element Elastic Model Reproducing Many of the Statistical Properties and Migration Patterns of Tremor. AGU Meeting, 2025.
2. Zhao, Z., **Xia, L.**, Zhu, W. and Yu, C. Seismic Tomography using Automatic Differentiation: Application to the San Francisco Bay Region. AGU Meeting, 2025.
1. **Xia, L.** and W. Zhu. Tomography of the Bay Area. “Earthquake of the Week” in UC Berkeley, 2023.

RESEARCH EXPERIENCE

Model slow slip events and tremor using elasto-frictional models

Supervisor: Prof. Allan M. Rubin

Princeton University

Sep. 2024 - Present

- Developed a discrete elastic model with static and sliding friction and a constant slip rate to simulate slow slip events.
- Characteristic slow events display the migration properties of elastic/friction models and stochastic properties of Brownian models.
- Used a simplified stochastic linear birth-death model to explain the observed proportionality between radiated energy rate and moment rate.

Use ADTomo to jointly inverse V_p and V_p/V_s ratio of the Bay Area

Supervisor: Prof. Weiqiang Zhu

UC Berkeley

Jul. 2023 - Oct. 2023

- Aimed to improve the previous velocity model in San Francisco Bay Area with more data coverage.
- Developed and modified scripts based on the fast-sweeping method, which does not assume that P-waves and S-waves follow the same path. Picked earthquake events for less computation cost and more uniform distribution. Utilized L-BFGS for optimization and obtained high-resolution models of V_p , V_s , and V_p/V_s ratio through joint inversion.
- Discovered a good correspondence between inversed velocity structures and the geological maps. Found narrow regions of high V_p/V_s ratios that might correspond to high pore pressure fault zones.

SELECTED AWARDS AND HONORS

Myhrvold-Havranek Graduate Fellowship, Princeton University	<i>May 2026</i>
Outstanding Undergraduate Graduate, USTC	<i>Apr. 2024</i>
Outstanding Undergraduate Graduation Thesis, USTC	<i>Jun. 2024</i>
China National Scholarship, Ministry of Education of China	<i>Oct. 2022, Sep. 2023</i>
Scholarship for Elite Class in Basic Disciplines (Class A), USTC	<i>Dec. 2022, Nov. 2023</i>

SKILLS

Programming: C/C++, Python, Julia, MATLAB, Mathematica, L^AT_EX, R

Tools: Git/GitHub, Unix Shell, SAC, GMT

TEACHING EXPERIENCE

Theoretical Mechanics A

USTC (*instructor: Prof. Gan Qin*)

Hefei, Anhui

Sept. 2022 - Jan. 2023

FIELD WORK EXPERIENCE

Geosciences Undergraduate Field Trip

Princeton University (*supervisor: Prof. Blair Schoene*)

Big Island, HI

May 2026

Physical Geology Field Practice

USTC (*supervisor: Prof. Jun Tang*)

Chaohu, Anhui

Jul. 2022